

Orchestrator Steps: `git submodule add docs_chain` into Lava

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Author: Stream C subagent (prep/analysis — read-only on parent; verified against upstream + gate scripts)
Classification: project-specific

1. Convergence Verification

Both mirrors report the same HEAD — §6.C satisfied before the add:

Mirror	SHA
git@github.com:vasic-digital/docs_chain.git HEAD	c606978ea702e72fd5f3a088cbb589cbea386b1f
git@gitlab.com:vasic-digital/docs_chain.git HEAD	c606978ea702e72fd5f3a088cbb589cbea386b1f

Pin to: c606978ea702e72fd5f3a088cbb589cbea386b1f

2. Parity-Presence Findings (at SHA c606978e)

Shallow-cloned from GitHub; HEAD verified = c606978e.

Artifact	Present?	Detail
helix-deps.yaml	YES	Schema v1, deps: [], leaf submodule
install_upstreams.sh	YES	Executable, --push / --dry-run flags
Upstreams/GitHub.sh	YES	UPSTREAMABLE_REPOSITORY=git@github.com:vasic-digital/docs_chain.git
Upstreams/GitLab.sh	YES	UPSTREAMABLE_REPOSITORY=git@gitlab.com:vasic-digital/docs_chain.git
CLAUDE.md	YES	

Artifact	Present?	Detail
		Opens with ## INHERITED FROM constitution/CLAUDE.md at line 3
AGENTS.md	YES	Opens with ## INHERITED FROM constitution/AGENTS.md at line 3
CONSTITUTION.md	YES	Opens with ## INHERITED FROM constitution/Constitution.md at line 3
QWEN.md	YES	Pointer file + ## INHERITED FROM constitution/CLAUDE.md marker
go.mod	YES	module digital.vasic.docs_chain,go 1.25.0

§6.C mirror convergence: CONFIRMED (both mirrors = c606978e)

§11.4.36 install_upstreams: PRESENT

§6.AD pointer-blocks: PRESENT in all three governance files

3. Mount Path Determination

Existing submodule naming observed in .gitmodules and submodules/:

- CamelCase subdirectories (physical dir): Auth, Cache, Challenges, Concurrency, Config, Containers, Database, Discovery, HelixQA, HTTP3, Mdns, Middleware, Observability, RateLimiter, Recovery, Security
- snake_case subdirectories: tracker_sdk
- HelixQA, HTTP3 use mixed-case/acronym convention
- .gitmodules path = values are ALL lowercase or snake_case:
submodules/containers, submodules/auth, submodules/tracker_sdk, submodules/helixqa

Conclusion: .gitmodules paths use lowercase/snake_case. Physical directory case is determined by git at add-time. For docs_chain (already snake_case in the GitHub repo name), the mount path is:

submodules/docs_chain

4. Orchestrator Command Sequence

Execute these commands **in order** from the Lava repo root. Do NOT run them out of order. The orchestrator owns the parent index — do not run these in parallel with any other parent-index-touching work.

Step 0 — Pre-flight: confirm convergence (already verified above, re-run for freshness)

```
# Re-confirm both mirrors report the same SHA
git ls-remote git@github.com:vasic-digital/docs_chain.git HEAD
git ls-remote git@gitlab.com:vasic-digital/docs_chain.git HEAD
# Both MUST print: c606978ea702e72fd5f3a088cbb589cbea386b1f
```

Step 1 — Add the submodule

```
git submodule add -b main \
  git@github.com:vasic-digital/docs_chain.git \
  submodules/docs_chain
```

This will:

- Create submodules/docs_chain/ populated from c606978e (current HEAD of main)
- Append the following stanza to .gitmodules:

```
[submodule "submodules/docs_chain"]
  path = submodules/docs_chain
  url = git@github.com:vasic-digital/docs_chain.git
  branch = main
```

- Stage .gitmodules and submodules/docs_chain in the parent index

Step 2 — Pin the submodule to the verified SHA

```
# The add checks out HEAD of main; pin explicitly to the verified
  SHA
git -C submodules/docs_chain checkout
  c606978ea702e72fd5f3a088cbb589cbea386b1f
# Re-stage the submodule pointer at the pinned SHA
git add submodules/docs_chain
```

Step 3 — Apply required governance-file patches in docs_chain (PRE-COMMIT FIX)

This step is MANDATORY before committing — see Gate-by-Gate table below; 5 gates will FAIL at pre-push if these fixes are not applied first.

In submodules/docs_chain/CLAUDE.md, replace the existing partial §6.S heading and add the missing §6.R and §6.X sections. The orchestrator should run scripts/inject-helix-inheritance-block.sh or apply the patches manually.

Patch A — submodules/docs_chain/CLAUDE.md (apply upstream in docs_chain, then bump pin):

Replace the existing §6.S section:

```
## §6.S – Continuation Document Maintenance (inherited)
```

With the exact heading the gate requires:

```
## §6.S – Continuation Document Maintenance Mandate (inherited  
2026-06-03, per §6.F)
```

(The body text See parent root `'/CLAUDE.md'` §6.S. ... is acceptable as-is — only the heading string is checked by the gate.)

Add immediately after the current §6.S section (or at end of file before any --- dividers):

```
## §6.R – No-Hardcoding Mandate (inherited 2026-05-06, per §6.F)
```

See root ``/CLAUDE.md`` §6.R. No connection address, port, header field name, credential, key, salt, secret, schedule, algorithm parameter, or domain literal in tracked source code. Every such value MUST come from ``.env`` (gitignored), generated config, runtime env var, or mounted file. Submodule MAY add stricter rules but MUST NOT relax.

```
## §6.X – Container-Submodule Emulator Wiring Mandate (inherited  
2026-05-13, per §6.F)
```

See root ``/CLAUDE.md`` §6.X. Every Android emulator instance the project depends on for testing MUST execute its emulator process INSIDE a podman/docker container managed by ``Submodules/Containers/``, NOT be host-direct-launched by Containers-submodule code that runs on the host. The Containers submodule's ``pkg/runtime/`` (rootless podman/docker auto-detection) brings the container up; ``pkg/emulator/`` orchestrates the AVD lifecycle inside it. Lava-side ``scripts/run-emulator-tests.sh`` is thin glue forwarding to the Containers CLI. The container-bound path is the gate — host-direct emulators are permitted for workstation iteration only. §6.X-debt tracks the wiring implementation owed to ``Submodules/Containers/``. This submodule MAY add stricter rules but MUST NOT relax.

Patch B — submodules/docs_chain/AGENTS.md — add at end:

§6.X – Container-Submodule Emulator Wiring Mandate (inherited 2026-05-13, per §6.F)

Inherited verbatim from parent Lava ``/CLAUDE.md`` §6.X. Every Android emulator instance MUST execute INSIDE a podman/docker container managed by ``Submodules/Containers/``. Host-direct emulator launches are permitted for workstation iteration only; the constitutional gate run (release tagging, real-device verification) MUST go through the container-bound path. ``pkg/runtime/`` brings the container up; ``pkg/emulator/`` orchestrates the AVD lifecycle inside it. §6.X-debt tracks the wiring implementation owed to the Containers submodule. This submodule MAY add stricter rules but MUST NOT relax.

Patch C — submodules/docs_chain/CONSTITUTION.md — add at end:

§6.X – Container-Submodule Emulator Wiring Mandate (inherited 2026-05-13, per §6.F)

See root ``/CLAUDE.md`` §6.X. Every Android emulator instance the project depends on for testing MUST execute its emulator process INSIDE a podman/docker container managed by ``Submodules/Containers/``, NOT be host-direct-launched by Containers-submodule code that runs on the host. The Containers submodule's ``pkg/runtime/`` (rootless podman/docker auto-detection) brings the container up; ``pkg/emulator/`` orchestrates the AVD lifecycle inside it. Lava-side ``scripts/run-emulator-tests.sh`` is thin glue forwarding to the Containers CLI. The container-bound path is the gate – host-direct emulators are permitted for workstation iteration only. §6.X-debt tracks the wiring implementation owed to ``Submodules/Containers/``. This submodule MAY add stricter rules but MUST NOT relax.

After applying patches upstream in docs_chain:

In the docs_chain repo:

```
git add CLAUDE.md AGENTS.md CONSTITUTION.md
git commit -m "fix(governance): add §6.R + §6.S exact heading +
§6.X mandate to CLAUDE/AGENTS/CONSTITUTION"
```

Required by Lava parent check-constitution.sh gates:

- §6.R gate: `grep -qF '## §6.R – No-Hardcoding Mandate'`
- §6.S gate: `grep -qF '## §6.S – Continuation Document Maintenance Mandate'`
- §6.X gate: `grep -qF '## §6.X – Container-Submodule Emulator Wiring Mandate'`

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```
git push github main
git push gitlab main
NEW_SHA=$(git rev-parse HEAD)
echo "New pin SHA: $NEW_SHA"
```

Then in the Lava parent:

```
git -C submodules/docs_chain checkout "$NEW_SHA"  
git add submodules/docs_chain
```

Step 4 — Update docs/CONTINUATION.md §3 pin index

Add the following line to the §3 "Submodule Pin Index" table in docs/CONTINUATION.md:

```
| docs_chain | submodules/docs_chain | `<NEW_SHA>` | main |  
| git@github.com:vasic-digital/docs_chain.git |
```

(Replace <NEW_SHA> with the actual SHA after the upstream fix commit in Step 3.)

Step 5 — Run the constitution gates dry-run (read-only verification)

```
bash scripts/check-constitution.sh  
bash scripts/check-canonical-root-and-upstreams.sh --strict
```

Both MUST exit 0 before proceeding.

Step 6 — Commit to parent

```
git add .gitmodules submodules/docs_chain docs/CONTINUATION.md  
git commit -m "feat(submodules): add docs_chain as submodules/  
docs_chain at <SHA>"
```

Pinned to SHA <SHA> – full parity with Lava parent gates:

- §6.AD pointer-blocks in CLAUDE.md, AGENTS.md, CONSTITUTION.md
- §11.4.36 install_upstreams.sh present
- helix-deps.yaml (leaf; deps: [])
- §6.R + §6.S + §6.X headings in all governance files
- §6.W: GitHub + GitLab mirrors both at same SHA (§6.C verified)

docs/CONTINUATION.md §3 pin index updated.

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Step 7 — Push to both mirrors

```
git push github master  
git push gitlab master
```

5. Resulting .gitmodules Stanza

After Step 1, .gitmodules will gain:

```
[submodule "submodules/docs_chain"]
  path = submodules/docs_chain
  url = git@github.com:vasic-digital/docs_chain.git
  branch = main
```

No git config -f .gitmodules tweaks required beyond what git submodule add -b main produces.

6. docs/CONTINUATION.md §3 Pin-Index Line

Draft line for the orchestrator to splice into the §3 table (update SHA after Step 3's fix commit):

```
| docs_chain | submodules/docs_chain | '<SHA-after-fix-commit>' |
main | git@github.com:vasic-digital/docs_chain.git |
```

7. Gate-by-Gate PASS / RISK Table

All gates run inside scripts/check-constitution.sh (invoked by .githubhooks/pre-push → scripts/ci.sh --changed-only).

scripts/check-constitution.sh gates

Gate	Script location	Requirement	docs_chain at c606978e	After upstream fix	Result
§6.R heading in submodules/*/CLAUDE.md	line 292–299	grep -qF '## §6.R – No–Hardcoding Mandate'	NOT present	After Patch A	RISK → PASS after fix
§6.S heading in submodules/*/CLAUDE.md	line 354–361	grep -qF '## §6.S – Continuation Document Maintenance Mandate'	Wrong: ... (inherited) not ... Mandate	After Patch A	RISK → PASS after fix
	line 392–399	grep -qF '## §6.X –	NOT present	After Patch A	

Gate	Script location	Requirement	docs_chain at c606978e	After upstream fix	Result
§6.X heading in submodules/*/CLAUDE.md		Container-Submodule Emulator Wiring Mandate'			RISK → PASS after fix
§6.X heading in submodules/*/AGENTS.md	line 392–399	same pattern	NOT present	After Patch B	RISK → PASS after fix
§6.X heading in submodules/*/CONSTITUTION.md	line 392–399	same pattern	NOT present	After Patch C	RISK → PASS after fix
§6.AD(4) pointer-block submodules/*/CLAUDE.md	line 490–503	grep -qE '^## INHERITED FROM constitution/'	PRESENT (line 3)	No change needed	PASS
§6.AD(4) pointer-block submodules/*/AGENTS.md	line 490–503	same	PRESENT (line 3)	No change needed	PASS
§6.AD(4) pointer-block submodules/*/CONSTITUTION.md	line 490–503	same	PRESENT (line 3)	No change needed	PASS
§6.N propagation (submodules/\$sm/CLAUDE.md)	line 192–209	iterates hardcoded 16-name list (no docs_chain)	n/a — not in list	Not applicable	NOT CHECKED — PASS (skip)
§6.O propagation	line 216–224	same 16-name list	n/a	Not applicable	NOT CHECKED — PASS (skip)
§6.P propagation	line 231–239	same 16-name list	n/a	Not applicable	NOT CHECKED — PASS (skip)
§6.Q propagation	line 246–254	same 16-name list	n/a	Not applicable	NOT CHECKED — PASS (skip)
	line 572–582			PASS	PASS

Gate	Script location	Requirement	docs_chain at c606978e	After upstream fix	Result
§6.W remote-host scan (submodules/*/)		no gitflic/ gitverse remotes	No remotes configured at add-time		
§11.4.6 no-guessing vocabulary	delegated to check-no-guessing-vocabulary.sh	scans .lava-ci-evidence/ sixth-law-incidents/ + crashlytics-resolved/ only	does not scan submodule content	Not applicable	PASS
§6.K containers emulator	line 146–169	conditional on submodules/ containers/ pkg/emulator/ existing	not about docs_chain	Not applicable	PASS
Credential scan (§6.H)	line 103–129	scans git ls-files — submodule content NOT in parent's git ls-files	not applicable	Not applicable	PASS

scripts/check-canonical-root-and-upstreams.sh gates

Gate	Requirement	docs_chain at c606978e	Result
§11.4.35(a) root CLAUDE.md + AGENTS.md inherit pointer	only checked for PARENT root (not submodules)	n/a	PASS (not checked for submodules)
§11.4.36 install_upstreams.sh in submodules/ docs_chain/	install_upstreams.sh OR variants present	install_upstreams.sh PRESENT at root	PASS

scan-no-hardcoded-ipv4.sh + scan-no-hardcoded-hostport.sh + scan-no-hardcoded-uuid.sh

These scanners explicitly exclude submodules/ via `grep -zVE '...|^submodules/|...'` — submodule content is excluded from the parent's git ls-files anyway (appears as a gitlink blob, not individual files).

Scanner	docs_chain impact	Result
scan-no-hardcoded-uuid.sh	Excluded — submodules/ path excluded	PASS
scan-no-hardcoded-ipv4.sh	Excluded — ^submodules/ in exclusion regex	PASS
scan-no-hardcoded-hostport.sh	Excluded — same exclusion	PASS

8. Summary: Pre-Add Actions Required

Before running `git submodule add`, the upstream docs_chain repo at c606978e has **5 gate failures** that will cause `check-constitution.sh` to exit 1 on the parent's first push:

1. **CLAUDE.md §6.R**: missing `## §6.R – No-Hardcoding Mandate` heading → add Patch A
2. **CLAUDE.md §6.S**: wrong heading (`...(inherited)` vs `...Mandate`) → fix in Patch A
3. **CLAUDE.md §6.X**: missing `## §6.X – Container-Submodule Emulator Wiring Mandate` → add Patch A
4. **AGENTS.md §6.X**: same missing heading → add Patch B
5. **CONSTITUTION.md §6.X**: same missing heading → add Patch C

Recommended flow: apply Patches A+B+C to docs_chain upstream, push to both mirrors, get the new SHA, THEN run `git submodule add` pointing at that new SHA.

Alternative (if orchestrator prefers one-shot): run `git submodule add` at c606978e, apply the patches to submodules/docs_chain/ locally, push the governance fixes to both docs_chain mirrors, bump the parent pin to the new SHA, commit once.

9. Forensic Note: §6.N, §6.O, §6.P, §6.Q NOT Required for docs_chain

The propagation checks for §6.N, §6.O, §6.P, §6.Q in `check-constitution.sh` iterate a **hardcoded 16-element list**

(Auth Cache Challenges Concurrency Config Containers Database Discovery HTTP3 Mdns Middleware Observability RateLimiter Recovery Security Tracker-SDK). docs_chain is not in this list. Therefore these four

propagation gates DO NOT apply to submodules/docs_chain/CLAUDE.md. No additions for §6.N/§6.O/§6.P/§6.Q are required. If the list is ever extended to include docs_chain in a future commit, those sections must be added at that time.